CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

Chapter 1: GST (Goods and Services Tax)

Topic-wise short notes with solved examples

GST is a single tax charged on the supply of goods and services. This chapter covers how the tax is worked out on a bill and how a dealer sets off the tax already paid.

1 What GST is

- **GST** is a tax charged on the value of goods and services each time they are supplied (sold).
- The amount on which the tax is calculated is the **taxable value**. Normally it is the selling price after any discount.
- The **rate of GST** depends on the goods or service and is always given in the question.

$$\text{GST} = \frac{\text{rate} \times \text{taxable value}}{100} \qquad \text{Amount of the bill} = \text{taxable value} + \text{GST}$$

- If a discount is given: taxable value = list price − discount. GST is charged on the discounted price.

2 Sales within a state: CGST and SGST

- When the seller and the buyer are in the **same state** (intra-state supply), the GST is split into two equal parts.
- **CGST** (Central GST) goes to the Central Government and **SGST** (State GST) goes to the State Government. In a Union Territory, UTGST takes the place of SGST.

$$\text{CGST} = \text{SGST} = \frac{1}{2} \times \text{GST}$$

- So at a GST rate of 18%, CGST is 9% and SGST is 9%.

Example 2.1 An article with list price ₹8,000 is sold at a discount of 10% within the same state. If the rate of GST is 18%, find the CGST, the SGST and the amount of the bill.

SOLUTION

Discount = 10% of ₹8,000 = ₹800, so taxable value = ₹7,200

CGST = 9% of ₹7,200 = ₹648 and SGST = 9% of ₹7,200 = ₹648

Amount of the bill = $7200 + 648 + 648$

CGST ₹648, SGST ₹648, bill amount ₹8,496

3 Sales between states: IGST

- When the seller and the buyer are in **different states** (inter-state supply), a single tax called **IGST** (Integrated GST) is charged at the full rate. It goes to the Central Government.

$$\text{IGST} = \text{GST at the full rate}$$

Example 3.1 A dealer in Delhi sells goods worth ₹15,000 to a dealer in Jaipur. If the rate of GST is 12%, find the tax charged and the amount of the bill.

SOLUTION

Delhi and Jaipur are in different states, so IGST is charged.

IGST $= 12\%$ of ₹15,000 $=$ ₹1,800

IGST ₹1,800; bill amount ₹16,800

4 A bill with several items

- Work out the taxable value and the GST for each item separately, because the rates can differ, and then add.

Example 4.1 Find the amount of this bill for a sale within the same state.

Item	Price per unit (₹)	Quantity	Rate of GST
A	500	2	12%
B	1,200	1	18%

SOLUTION

Item A: taxable value $= 500 \times 2 =$ ₹1,000; GST $= 12\%$ of 1,000 $=$ ₹120 (CGST ₹60, SGST ₹60)

Item B: taxable value $=$ ₹1,200; GST $= 18\%$ of 1,200 $=$ ₹216 (CGST ₹108, SGST ₹108)

Total taxable value $=$ ₹2,200 and total GST $=$ ₹336

Amount of the bill $=$ ₹2,536

5 Finding the price before tax

- If the amount of the bill and the rate $r\%$ are known:

$$\text{Taxable value} = \frac{\text{amount of the bill} \times 100}{100 + r}$$

Example 5.1 The amount of a bill, including GST at 18%, is ₹5,310. Find the price of the article before tax.

SOLUTION

Let the taxable value be ₹x. Then $x + \frac{18}{100}x = 5310$, so $\frac{118x}{100} = 5310$

$$x = \frac{5310 \times 100}{118}$$

The price before tax is ₹4,500.

6 Input tax credit

- A dealer pays GST when buying goods (**input GST**) and collects GST when selling them (**output GST**).
- The dealer does not pay tax twice. The tax already paid on purchases is set off against the tax collected on sales. This set-off is the **input tax credit (ITC)**.

$$\text{GST payable to the government} = \text{output GST} - \text{input GST}$$

- So each dealer effectively pays tax only on the value that he or she adds.

Example 6.1 A dealer buys an article for ₹20,000 and sells it to a consumer for ₹25,000, both within the same state. The rate of GST is 18%. Find (i) the input CGST and SGST (ii) the output CGST and SGST (iii) the net GST payable by the dealer (iv) the price paid by the consumer.

SOLUTION

(i) Input CGST $= 9\%$ of 20,000 $=$ ₹1,800 and input SGST $=$ ₹1,800

(ii) Output CGST $= 9\%$ of 25,000 $=$ ₹2,250 and output SGST $=$ ₹2,250

(iii) Net CGST $= 2250 - 1800 =$ ₹450 and net SGST $=$ ₹450, so net GST $=$ ₹900

(iv) Price paid by the consumer $= 25000 + 2250 + 2250$

Net GST payable ₹900 (CGST ₹450 + SGST ₹450); the consumer pays ₹29,500.

Example 6.2 A dealer in Mumbai buys goods for ₹10,000 from a dealer in Delhi and sells them for ₹12,000 to a customer in Mumbai. The rate of GST is 18%. Find the net GST payable by the Mumbai dealer.

SOLUTION

Purchase from another state: input IGST $= 18\%$ of 10,000 $=$ ₹1,800

Sale within the state: output GST $= 18\%$ of 12,000 $=$ ₹2,160 (CGST ₹1,080 + SGST ₹1,080)

Net GST payable $= 2160 - 1800$

₹360

REMEMBER

Check: the dealer added ₹2,000 of value, and 18% of ₹2,000 is ₹360.

7 Common mistakes

- Charging GST on the list price when a discount is given. Tax is on the discounted price.
- Charging both CGST and SGST at the full rate. Each is half the GST rate.
- Using CGST and SGST for a sale between two states. That sale carries IGST.
- Forgetting to subtract the input tax when the net tax payable is asked.